

Performance Testing

Date	15 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual Testing, Flask Development Server
Type of Testing	Load Testing
Target Module / API	Crop Prediction Module (/predict)
Test Environment	Local System (Windows 11, Python 3.12, Flask)
Test Date	03-07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits crop prediction	1	30	Prediction displayed within 2 seconds
2	Multiple users accessing prediction page	10	60	Server responds without errors
3	Continuous prediction requests	20	120	Stable response and correct predictions

4	Invalid input submission	5	30	Error handled gracefully without crashing
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.3 seconds	Pass	Fast response
2	Response Time (Max)	< 5 seconds	3.2 seconds	Pass	Within acceptable limit
3	Throughput (Req/sec)	> 10 Req/sec	18 Req/sec	Pass	Good request handling
4	Error Rate	< 1%	0%	Pass	No request failures
5	CPU Utilization	< 80%	42%	Pass	CPU usage remained stable
6	Memory Utilization	< 80%	48%	Pass	Memory usage acceptable

Step 4: Observations & Analysis

Key Findings:

- Crop prediction requests were processed quickly with low response time.
- The application remained stable under multiple user requests.
- Machine learning predictions were generated accurately without delays.

Bottlenecks Identified:

- Performance may reduce with a very large dataset.
- Flask development server is not suitable for production deployment.
- Prediction speed depends on system hardware.

Optimization Steps Taken:

- Used a pre-trained Pickle (.pkl) model for faster predictions.
- Optimized data preprocessing before prediction.
- Implemented input validation and exception handling to reduce errors.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

